

Qns	Answer	Marks
2(a)(i)	Total momentum of an isolated system of interacting bodies before and after collision remains constant if no net external force acts on the system.	1
(a)(ii)	Presence of external force like weight / force from the ground	1
(b)	By Conservation of Energy, loss of gravitational potential energy = gain in kinetic energy $mgh = \frac{1}{2}mv^2$ $v = \sqrt{2gh}$ $= \sqrt{2(9.81)(2.0)}$ $= 6.26 \text{ m s}^{-1}$	1
(c)	collision is elastic, total KE before and after bounce conserved $\frac{1}{2}(0.8)(6.26)^2 = \frac{1}{2}(0.3)(v^2) + \frac{1}{2}(0.5)(3.2)^2$ $v = 9.35 \text{ m s}^{-1}$	1 1

Qns	Answer	Marks
2(e)(i)	Since the direction of field strength is directed upwards at point O, the	